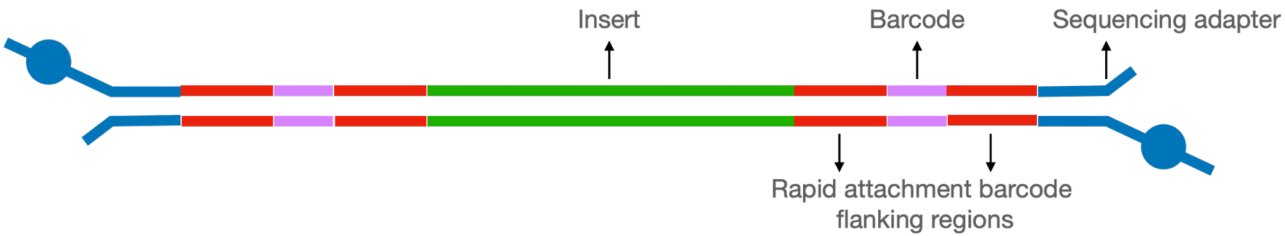
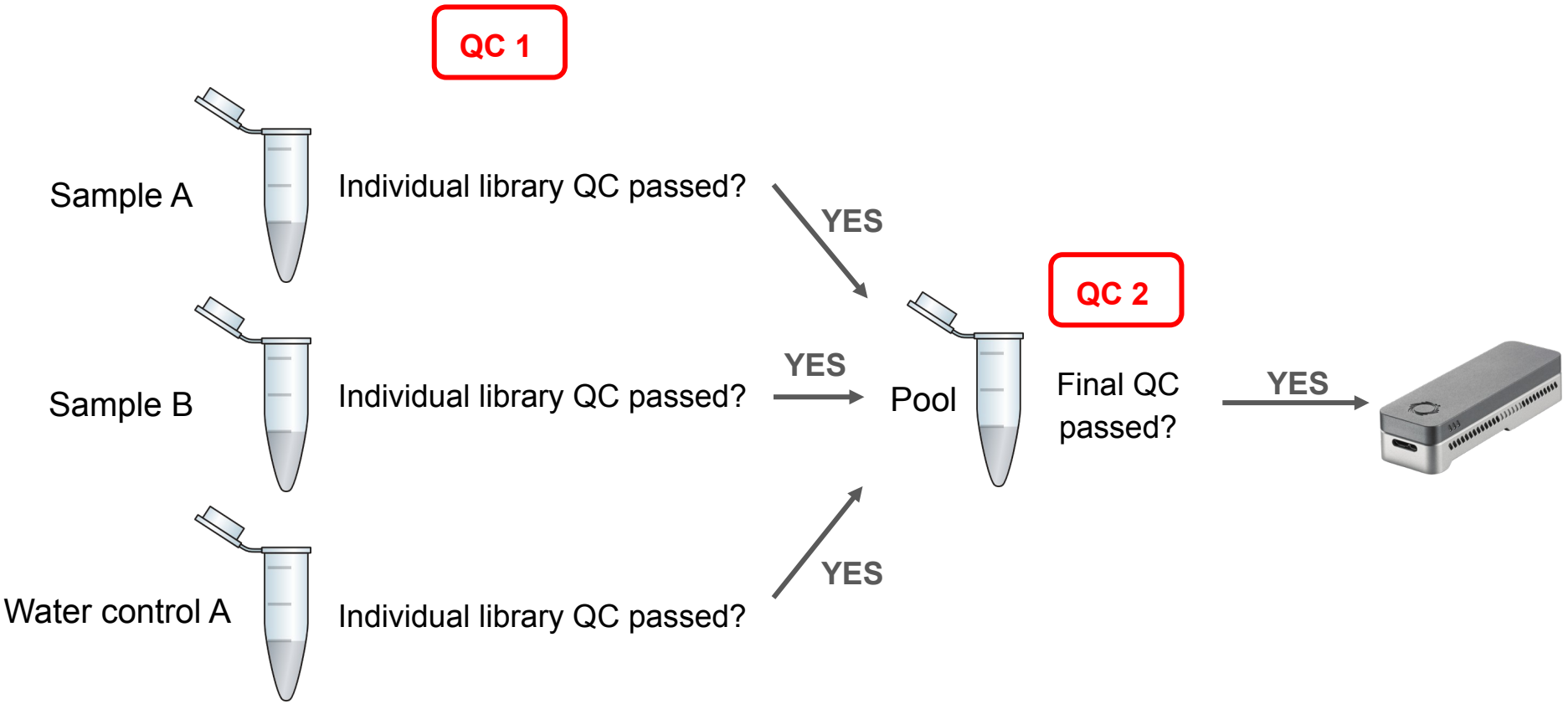
Library QC for ONT metagenomics

Library Quality Control (QC)

- The purpose of QC is to ensure that there is sufficient and good quality library for sequencing.
- Quantity (concentration)
  - Amount of library is measured using Qubit (or other fluorescence based method)
- Quality (length)
  - Measured using capillary electrophoresis (Tapestation, Bioanalyzer, Fragment Analyzer)
  - If capillary electrophoresis is unavailable, regular gel electrophoresis may be used with a diagnostic PCR

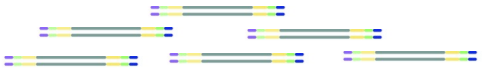


Overview of Library QC

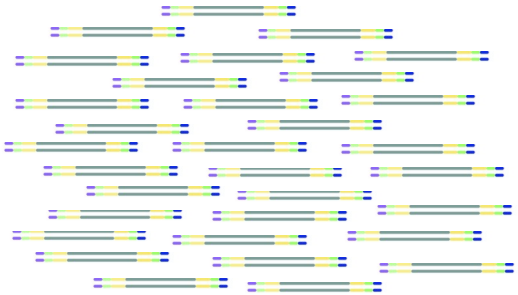


Library prep QC

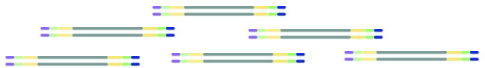
Concentration



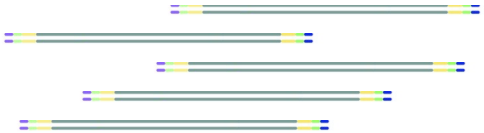
vs



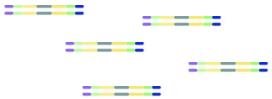
Length



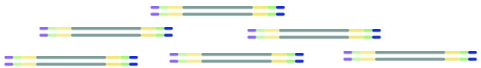
vs



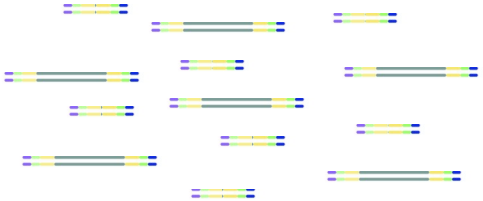
or



Remove adapter dimers



vs



Assessing concentration is important for even pooling and accurate loading

Why is an accurate concentration important?

- **Individual samples:** to pool equal amounts of different samples together
  - Aim for an even amount of sequencing data per sample

- **Final library:** The total amount of final library needs to be within a specific range*
  - For the MinION 10.4 flow cell, this range is 10-50 fmol
    - **We have found that 100-150 fmol works best**

*Nanopore is more forgiving than Illumina in accommodating a wider range of input loading amounts and we find that loading more than 50 fmol works well

Library concentration is best assessed using fluorometry like Qubit

Fluorometry
(e.g. Qubit)



✓ **Use for concentration**

- Detects fluorescent dyes that only bind DNA

Library concentration is best assessed using fluorometry like Qubit

Fluorometry
(e.g. Qubit)

✓ **Use for concentration**



- Detects fluorescent dyes that only bind DNA

Electrophoresis
(e.g. QSep, Tapestation, Bioanalyzer, gel)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

✓ **Use for size profiling (“quality”) and molar concentrations (for pooling)**

Bioanalyzer or Tapestation may be used for pooling, if necessary

Fluorometry
(e.g. Qubit)

✓ **Use for concentration**



- Detects fluorescent dyes that only bind DNA

Electrophoresis
(e.g. QSep, Tapestation, Bioanalyzer, gel)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

✓ **Use for size profiling (“quality”) and molar concentrations (for pooling)**

Nanodrop is not recommended for assessing concentration

Spectrophotometry
(e.g. Nanodrop)



- DNA absorbs light most strongly at 260 nm
- Level of absorbance ~ concentration of DNA

BUT other molecules (RNA, proteins) & contaminants also absorb light at 260 nm, causing up to 10-fold **overestimation** of the DNA concentration!

✗ **Do not use for concentration**

✓ **Use for detecting contaminants**

How to use the Qubit

Qubit dsDNA High Sensitivity (HS) Assay kit



- dsDNA HS Reagent (200x concentrate)
- dsDNA HS Buffer
- dsDNA HS Standard #1
- dsDNA HS Standard #2

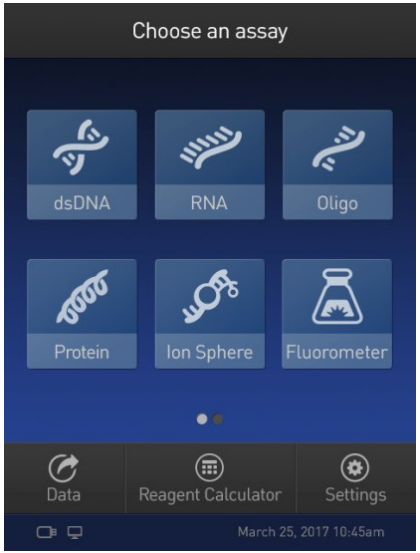
1. Prepare Working Solution:
   - Diluting HS reagent 1:200 in HS buffer

1. Dilute in Working Solution:
   - Standards: 10 µl standard + 190 µl Working Solution
   - Samples: 1 µl sample + 199 µl Working Solution

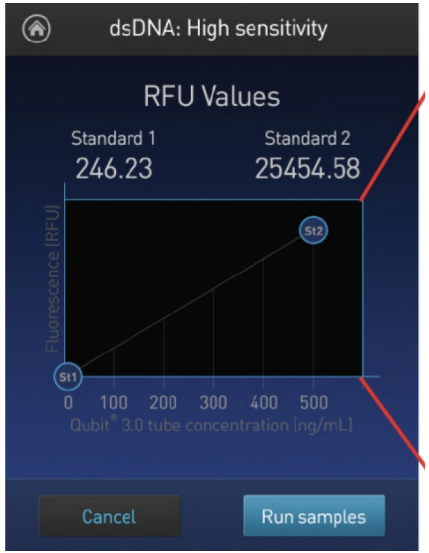
1. Vortex 2-3 seconds before running standards/samples

How to use the Qubit

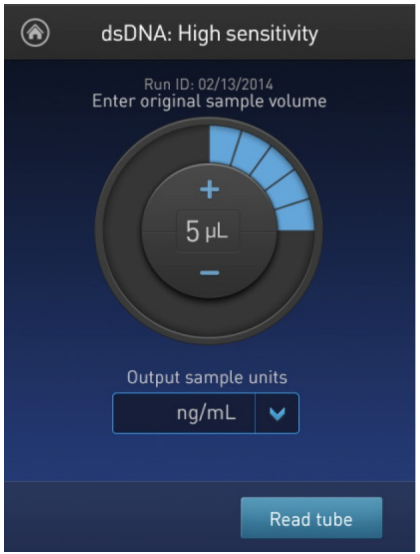
Qubit dsDNA High Sensitivity (HS) Assay kit



Select dsDNA HS workflow



Run the 2 standards



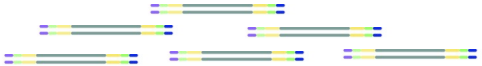
Enter sample volume added to tube



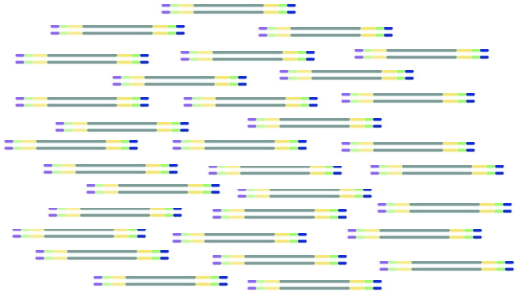
Run sample
Top value = original conc.
Bottom value = diluted conc.

Library prep QC

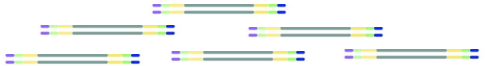
Concentration



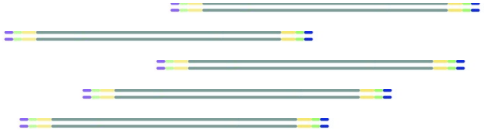
vs



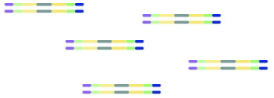
Length



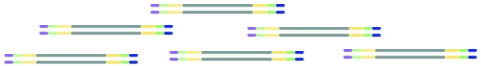
vs



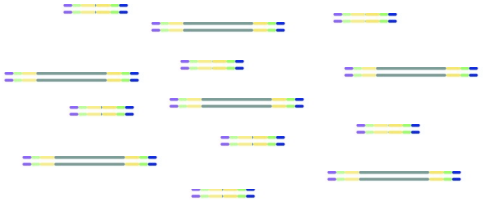
or



Remove adapter dimers



vs



Assessing length can provide insight to the quality of the library:

Is the library good enough to load?

Why is it important to assess library size distribution?

- Depending on sample quality, there can be huge variations in the library size (100s of bases to 1000s of bases)

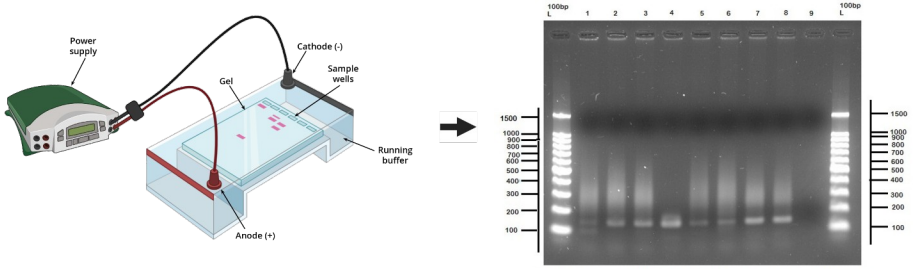
- Non-specific products can sometimes be formed
  - Primer/oligo dimers when high input cDNA concentrations are used
  - Long concatemers that can block pores and reduce sequencing output (~10kb or more)

- Fragment length distribution impacts molarity calculations for pooling and sequencer loading
  - Convert from ng/µl (Qubit) + Length bp to nM (fmol/µL)

Library size is assessed with electrophoresis methods

How to assess library size distribution?

Gel electrophoresis



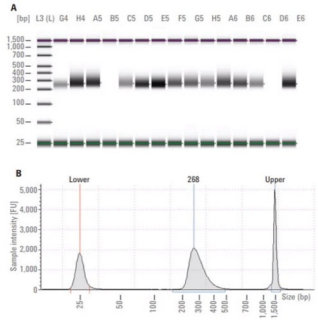
1. PCR amplify an aliquot of the library using “universal” primers
2. Gel electrophoresis of ladder and samples

- Takes ~ 4 hrs
- Does not require specialized equipment
- Tends to slightly overestimate library size

Capillary electrophoresis

E.g. Tapestation, Bioanalyzer





1. Add buffer to aliquot of library
2. Run machine

- Tapestation ~ 10-20 minutes
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

Diagnostic PCR method

1. Take 1 uL of the individual libraries or final pool and perform a standard PCR reaction for 20 cycles
   Note: the PCR product will only be used for diagnostic QC, not for sequencing

1. For each sample or pool, setup the amplification reaction as follows:

Component	Volume (µL)
LongAmp Master Mix (2x)	25
Nuclease-free Water	23.5
RLB-PCR Primer* (10µM)	0.5
DNA Library Template	1
Total Volume	50

*This primer is only for diagnostic PCR

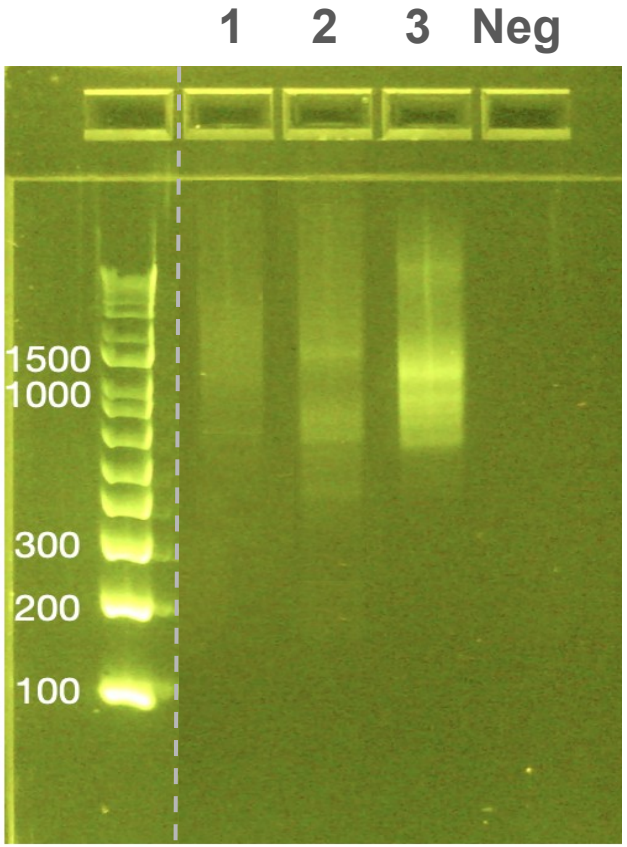
3. Use the following cycling conditions:

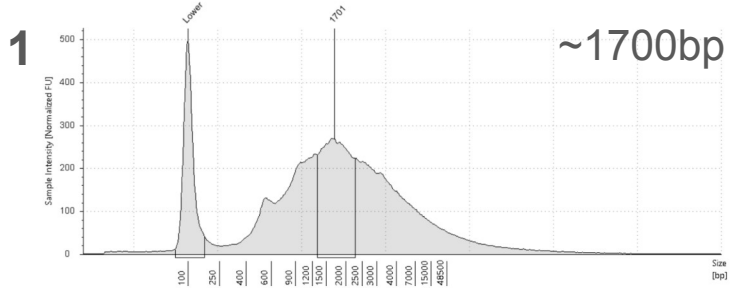
Thermocycler (lid at 105˚C)	Cycles
95 °C for 45s	1
95 ˚C for 15s	20
56 ˚C for 15s	
65 °C for 5 min	
65˚C for 10 mins	1
Hold at 10˚C	∞

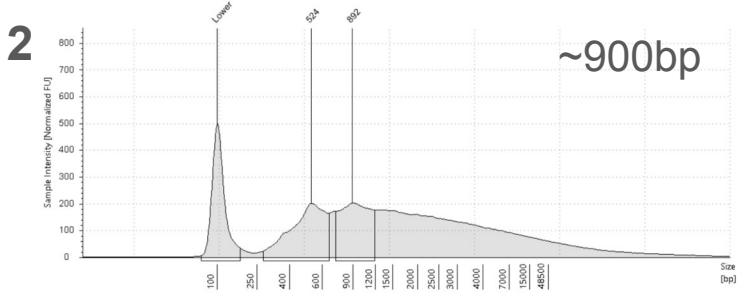
4. Run PCR amplified product on a gel to visualize and estimate the library size.

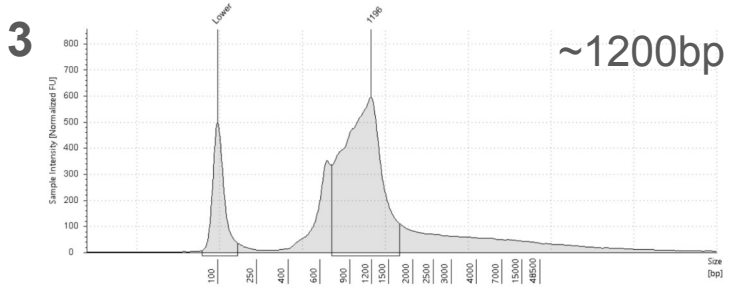
CZ Biohub Protocols: Nanopore mNGS

Diagnostic PCR results compared to Tapestation





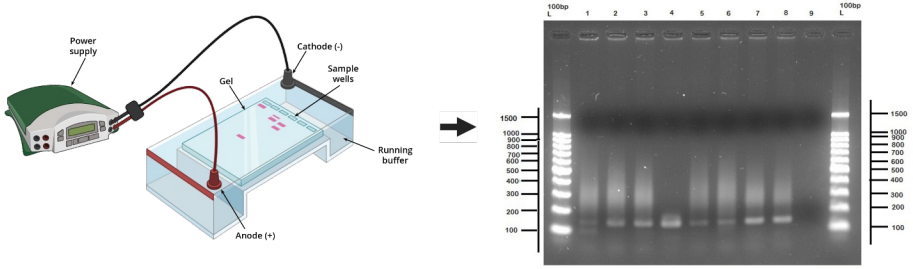




Library size is assessed with electrophoresis methods

How to assess library size distribution?

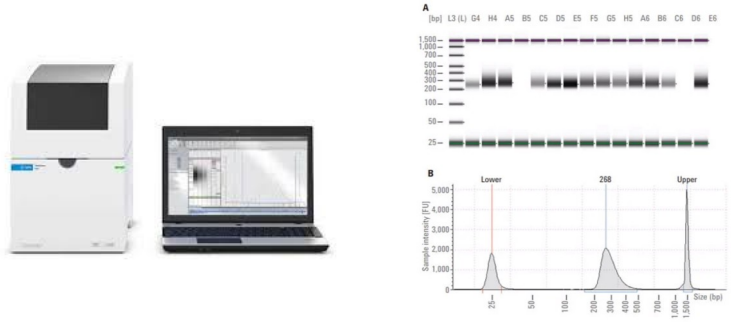
Gel electrophoresis



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   - Takes ~ 4 hrs
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Capillary electrophoresis

E.g. Tapestation, Bioanalyzer



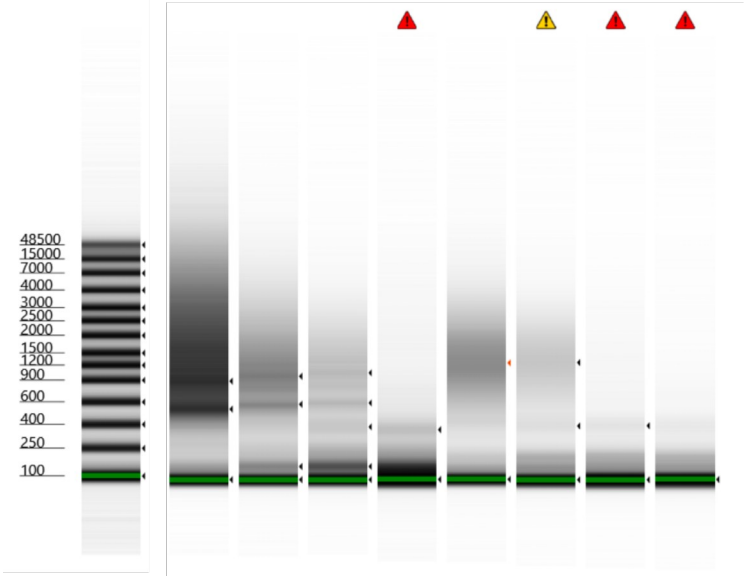
1. Add buffer to aliquot of library
2. Run machine

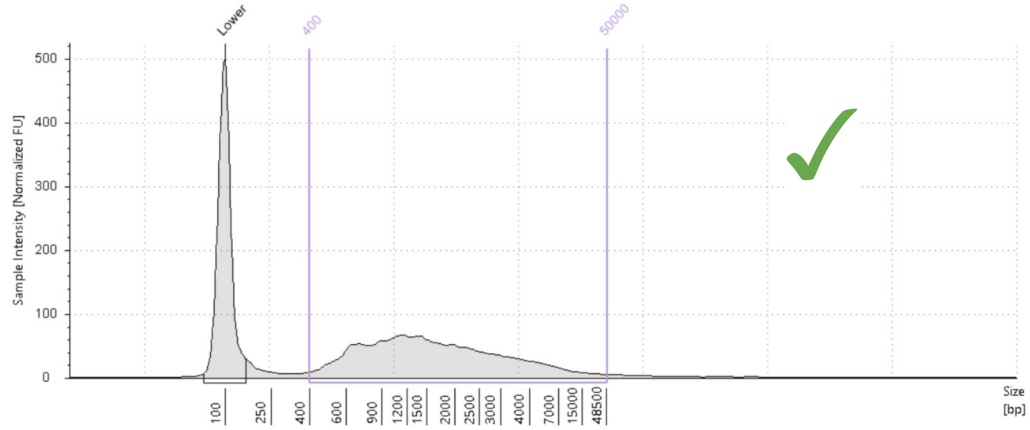
- Tapestation ~ 10-20 minutes
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

Tapestation genomic DNA kit can be used to assess library size

Tapestation

- Genomic DNA kit is most suitable since fragments can be 1000s of bases long





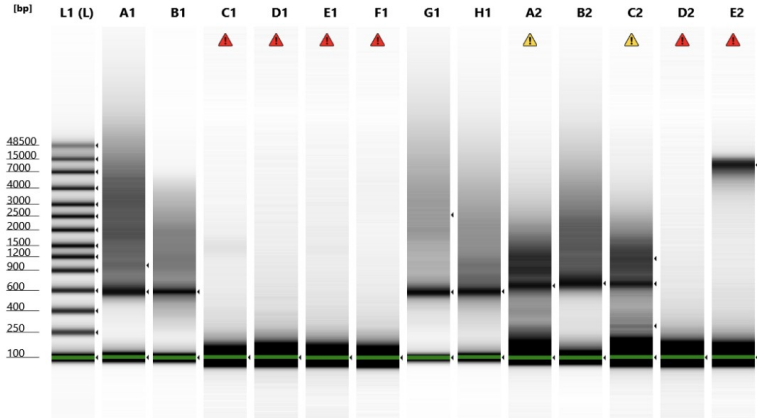
Region Table

From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color
400	50000	2379	12.2	15.0	90.56		

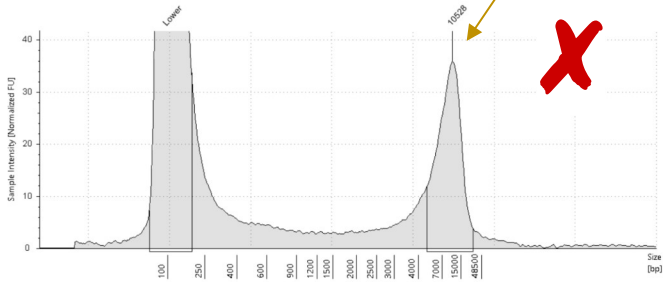
Look out for ~10kb concatemer artifacts - they will clog the flow cell pores

Tapestation

- Genomic DNA kit is most suitable since fragments can be 1000s of bases long



Concatemer peak at ~10kb, with no other peaks*



Sample Table

Well	DIN	Conc. [ng/µl]	Sample Description	Alert	Observations
E2	-	2.39	L13	▲	Sample concentration outside functional range for DIN and the assay

*Consider the input going in, and the library size you would expect. In this case, a 10kb peak was seen for RNA input, which is much longer than the fragment size we'd expect. It turns out that the 10kb peak was a concatemer artifact created during barcoding PCR.

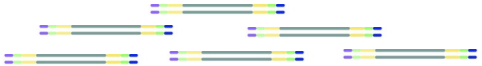
Library size

- Checking library size is important for troubleshooting
- It is recommended to routinely check library size when establishing the protocol or expanding to new sample types
- If results are consistent and sample type / source are constant, it is possible for the library size to be estimated

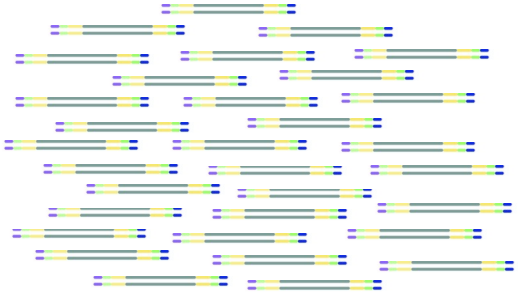
- Size may be evaluated by:
  - Diagnostic PCR + gel electrophoresis
  - QSep / Tapestation / Bioanalyzer / Fragment analyzer
- Size will be used for concentration calculation

Library prep QC

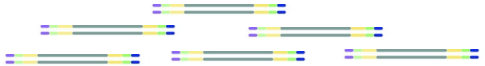
Concentration



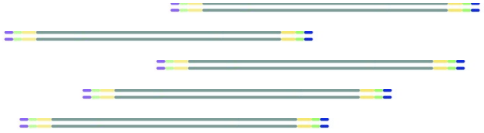
vs



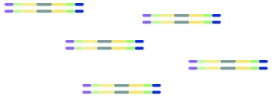
Length



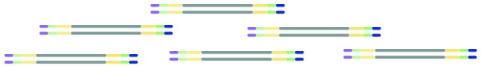
vs



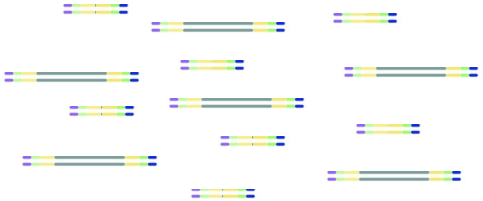
or



Remove adapter dimers

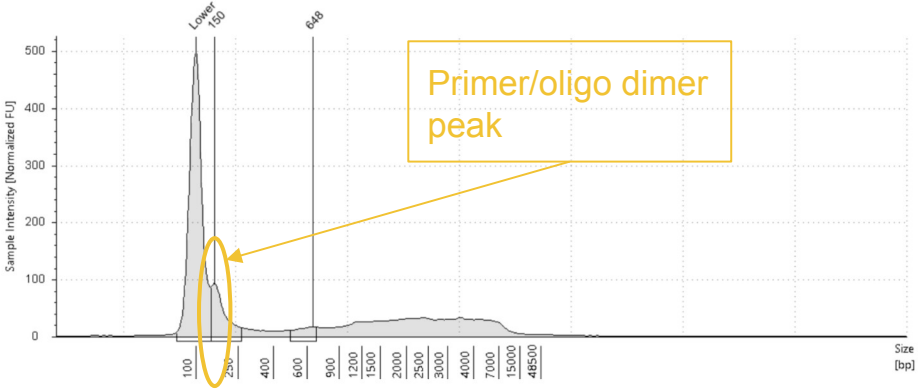


vs



Library prep QC - Clean up primer/oligo dimers

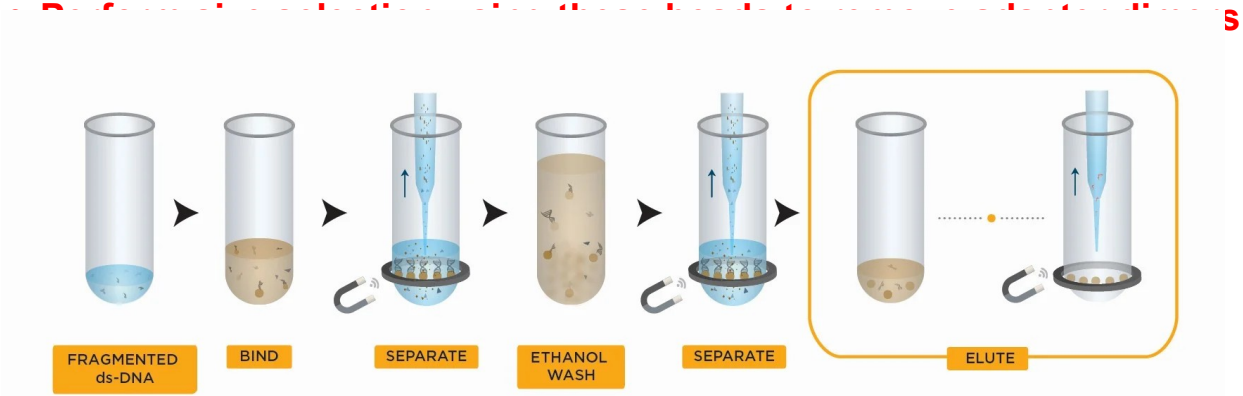
- Primer/oligo dimers:
  - Occur when primers and unused 9N and TSO oligos interact during the PCR amplification
  - Will not be informative if sequenced



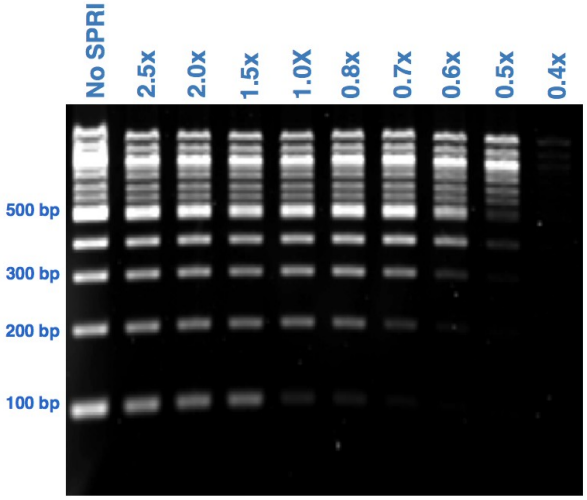
SPRI beads may be used to remove adapters and reagents

SPRIselect or AMPure XP SPRI select beads (paramagnetic)

- The longer a fragment, the better it will bind to the beads
- The fewer beads you add, the more short fragments you'll lose
- Perform size selection using these beads to remove adapter dimers



1. Bind DNA to magnetic beads; 2. Separate beads from contaminants;
3. Wash magnetic beads with 85% ethanol to remove contaminants;
4. Elute DNA from magnetic beads; 5. Transfer sample to new tube



$$\text{SPRI ratio} = \frac{\text{Volume beads}}{\text{Volume reaction}}$$

Example: for a 1.8x SPRI clean-up:

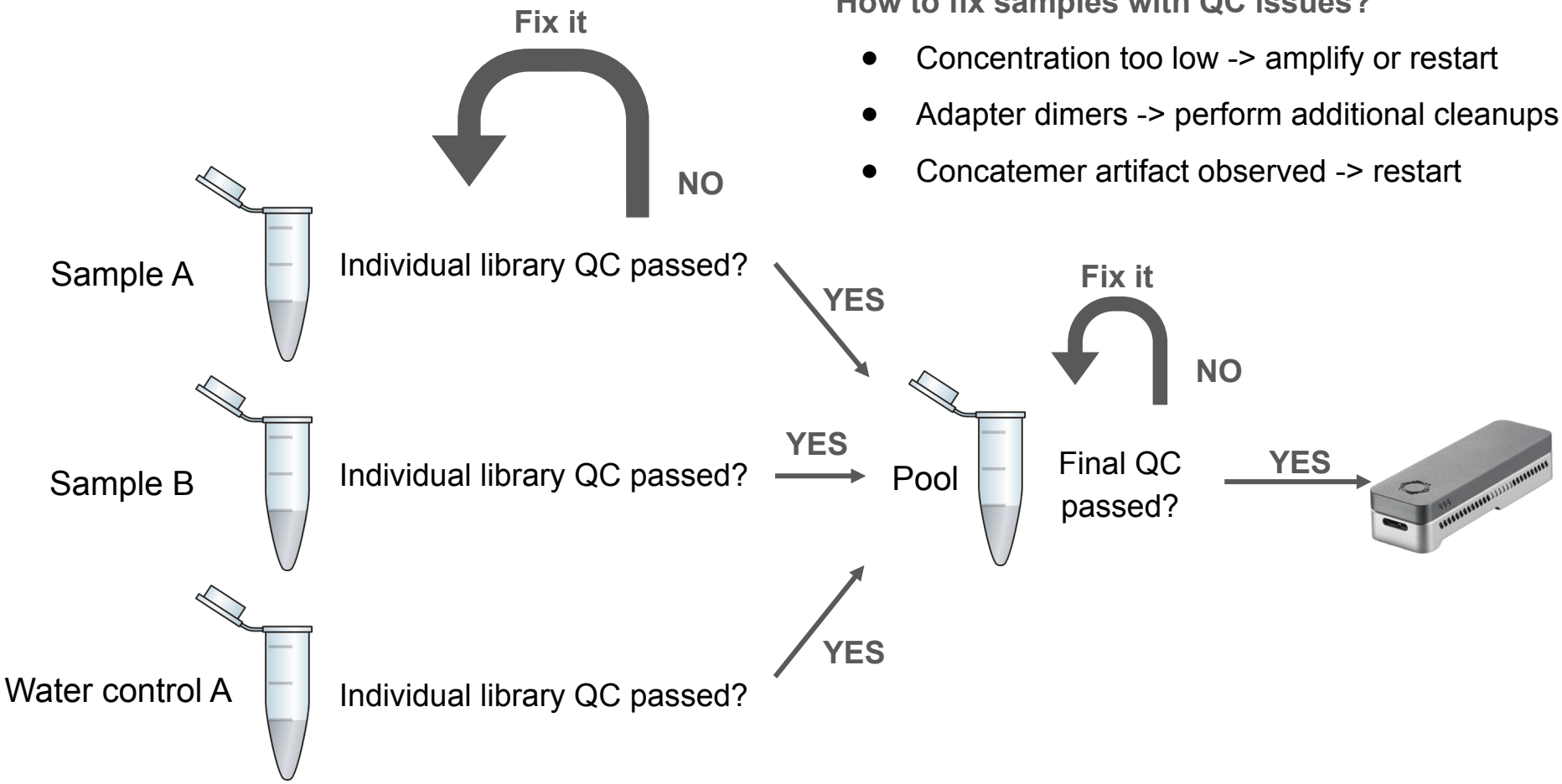
- Add 18 µl of SPRI beads to 10 µl of sample

Library prep QC - Clean up adapter dimers

AMPure XP SPRI select beads (paramagnetic)

- Size selection using SPRI beads to remove adapter dimers
- Samples can first be pooled and then a 0.6x SPRI clean-up can be performed

Library prep QC



How to fix samples with QC issues?

- Concentration too low -> amplify or restart
- Adapter dimers -> perform additional cleanups
- Concatemer artifact observed -> restart

How much concentration do I really need?

- Final pool
  - Need 100-150 fmol to load the flow cell
  - = 11ul of 10-15nM

- Individual samples:
  - Ideally individual samples would also have 10-15nM
    - (7-10ng/ul for 1000bp library)
    - (13-20ng/ul for 2000bp library)
  - Total mass (ng) per sample added to pool will depend on:
    - library size
    - # of samples in the pool

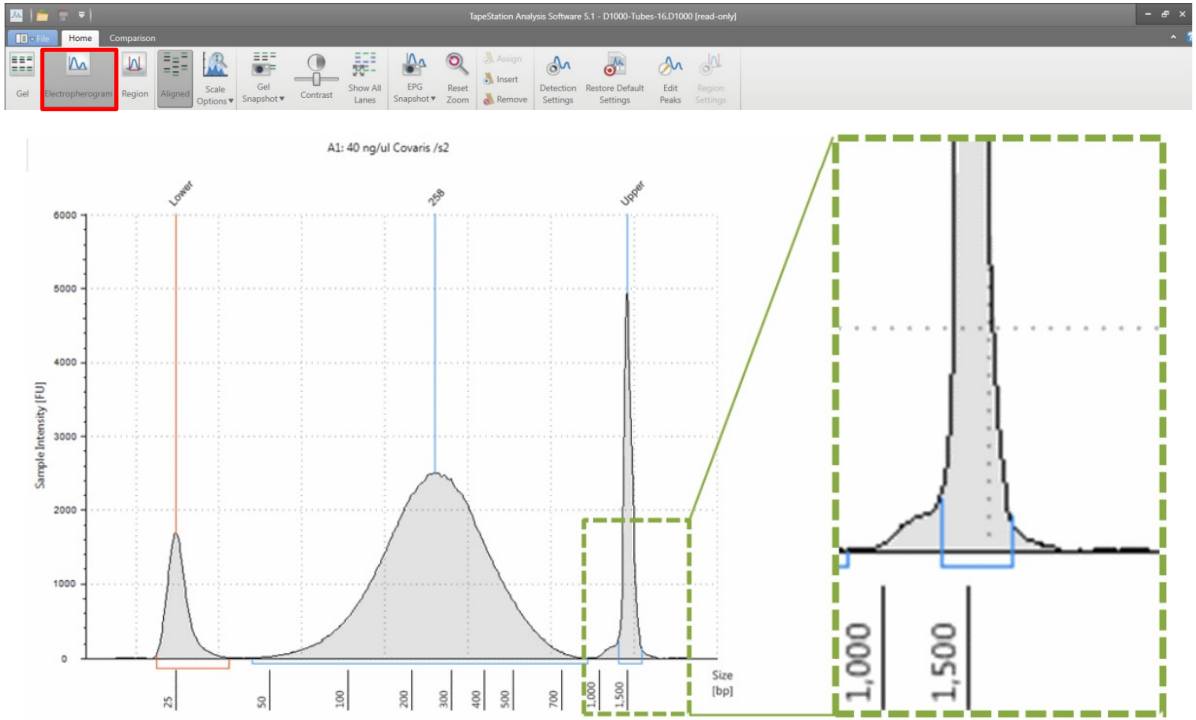
Using the Tapestation

Choosing the right assay according to your starting material

ScreenTape Assay	Quick Guide Number	
Agilent D1000 ScreenTape Assay	G2991-90031	
Agilent High Sensitivity D1000 ScreenTape Assay	G2991-90131	
Agilent RNA ScreenTape Assay	G2991-90021	
Agilent High Sensitivity RNA ScreenTape Assay	G2991-90121	RNA 500 – 10000 pg/µL
Agilent Genomic DNA ScreenTape Assay	G2991-90041	DNA 200 to >60000 bp, 10 – 100 ng/µL
Agilent D5000 ScreenTape Assay	G2991-90051	
Agilent High Sensitivity D5000 ScreenTape Assay	G2991-90151	Sequencing libraries 100 - 5,000 bp, 10 - 1,000 pg/µL
Agilent Cell-free DNA ScreenTape Assay	G2991-90060	

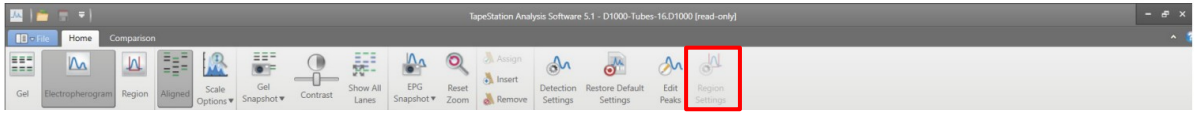
Note, sensitivity is guided by input concentration.

How to analyse a TapeStation run?

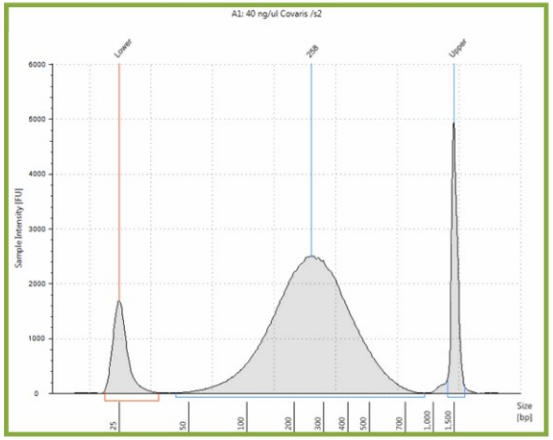


1. Ensure that all marker peaks (‘Lower’ and ‘Upper’) are properly assigned. **Concentration values are calculated using the area of the sample peak compared to the known concentration of the upper marker.**

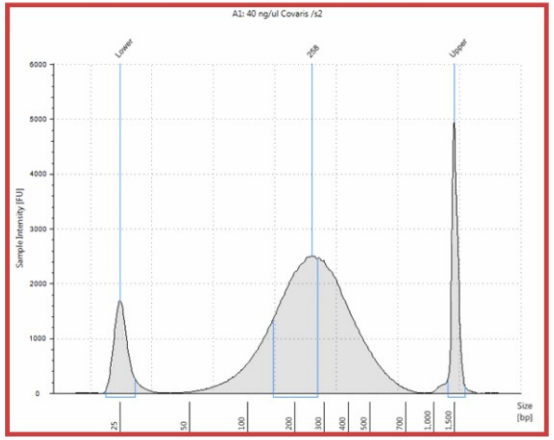
How to analyse a TapeStation run?



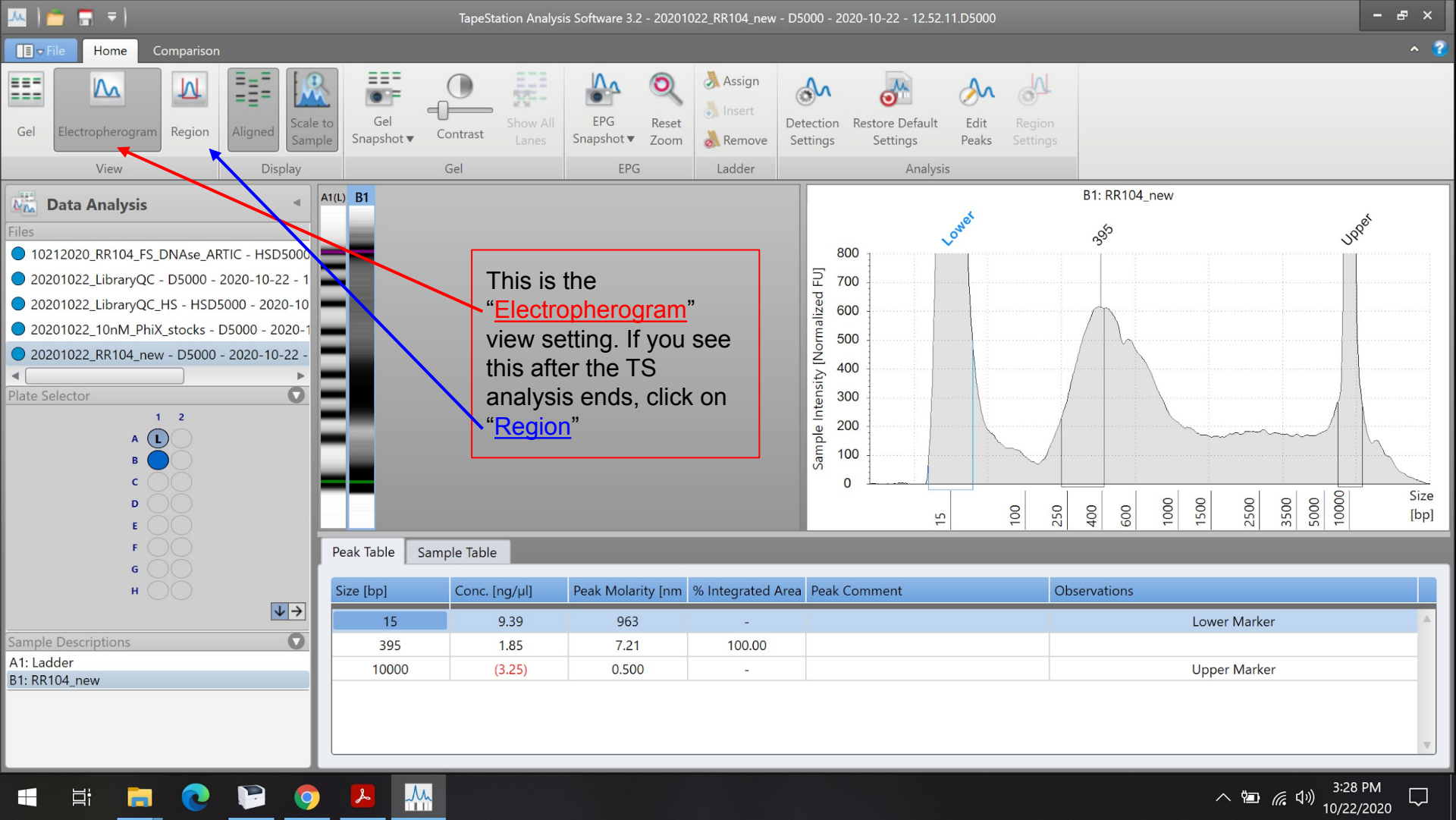
Correct peak integration
Concentration: 41.2 ng/µL



Incorrect peak integration
Concentration: 14.9 ng/µL



1. Ensure that all marker peaks (‘Lower’ and ‘Upper’) are properly assigned. **Concentration values are calculated using the area of the sample peak compared to the known concentration of the upper marker.**
2. Ensure that sample peaks are properly integrated by manually adjusting the peak when necessary.


TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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20201022_RR104_new - D5000 - 2020-10-22 -
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
A1(L)
B1
This is the "Electropherogram" view setting. If you see this after the TS analysis ends, click on "Region"
B1: RR104_new
Lower
395
Upper
Sample Intensity [Normalized FU]
Size [bp]
Peak Table
Sample Table
Size [bp]
Conc. [ng/µl]
Peak Molarity [nm
% Integrated Area
Peak Comment
Observations
15
9.39
963
-
Lower Marker
395
1.85
7.21
100.00
10000
(3.25)
0.500
-
Upper Marker
3:28 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000

File | Home | Comparison

Gel | Electropherogram | Region | Aligned | Scale to Sample | Gel Snapshot | Contrast | Show All Lanes | EPG Snapshot | Reset Zoom | Assign | Insert | Remove | Detection Settings | Restore Default Settings | Edit Regions | Region Settings

View | Display | Gel | EPG | Ladder | Analysis

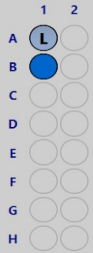
Click on “Region Settings”

Data Analysis

Files

- 10212020_RR104_FS_DNAse_ARTIC - HSD5000
- 20201022_LibraryQC - D5000 - 2020-10-22 - 1
- 20201022_LibraryQC_HS - HSD5000 - 2020-10
- 20201022_10nM_PhiX_stocks - D5000 - 2020-1
- 20201022_RR104_new - D5000 - 2020-10-22 -

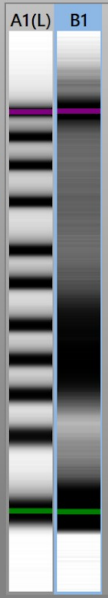
Plate Selector

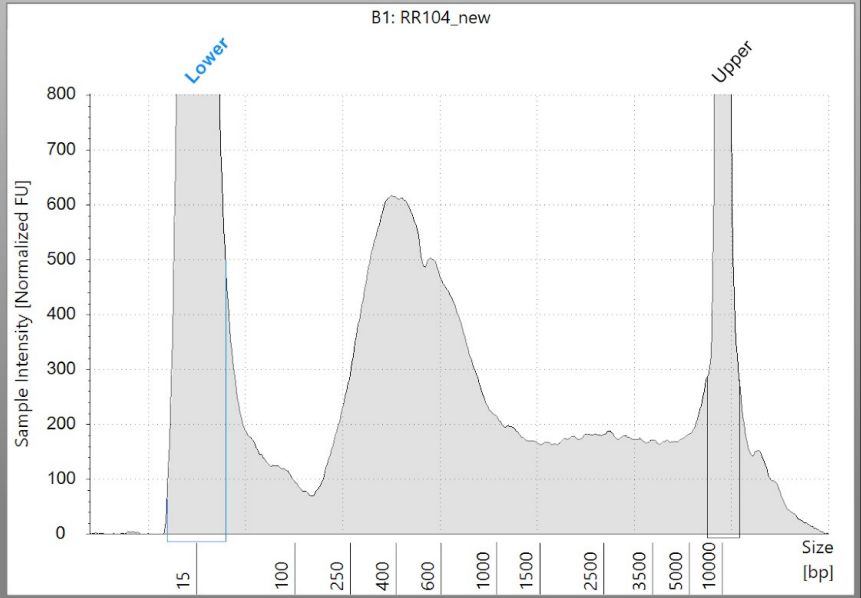


Sample Descriptions

A1: Ladder

B1: RR104_new

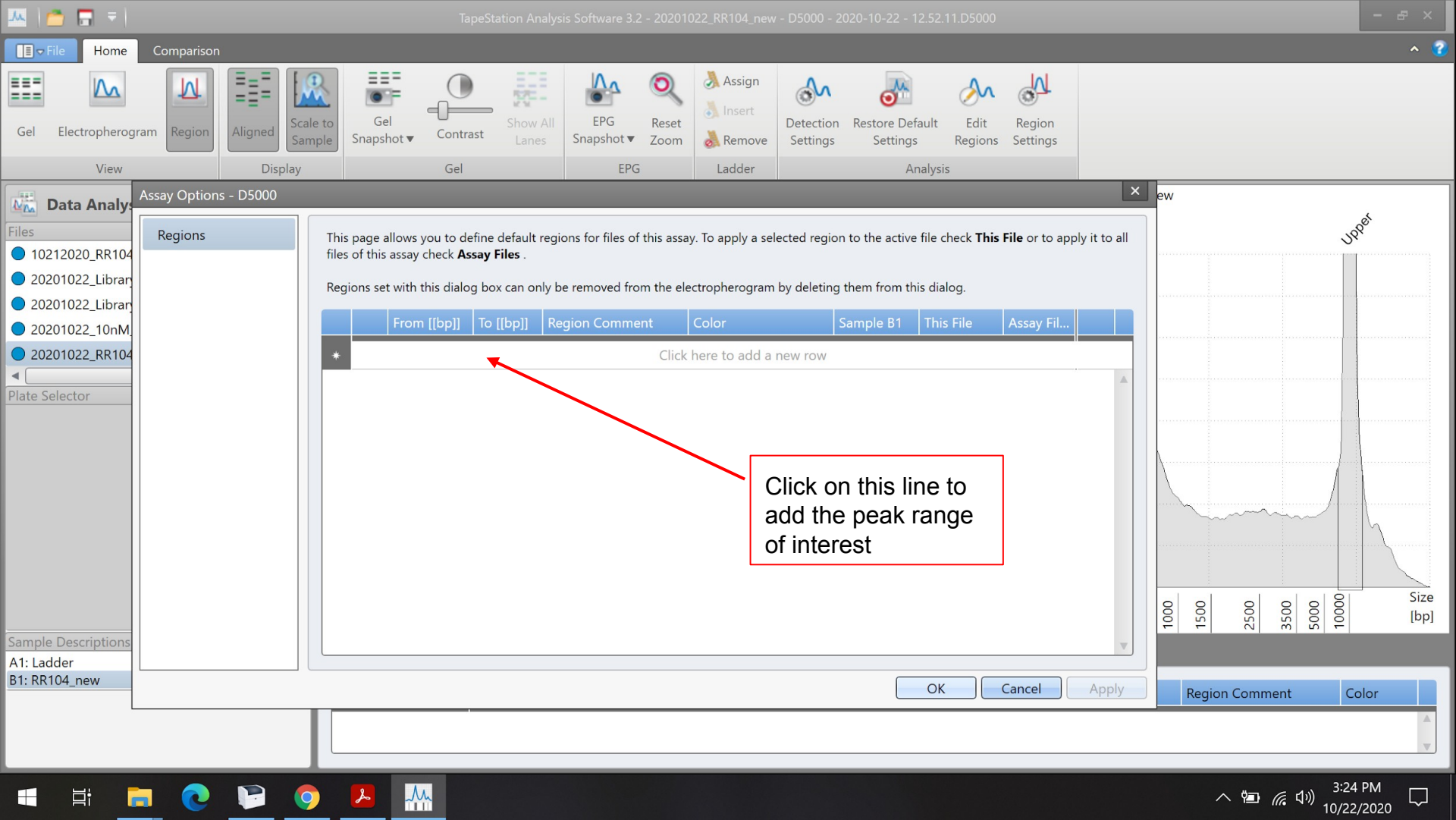


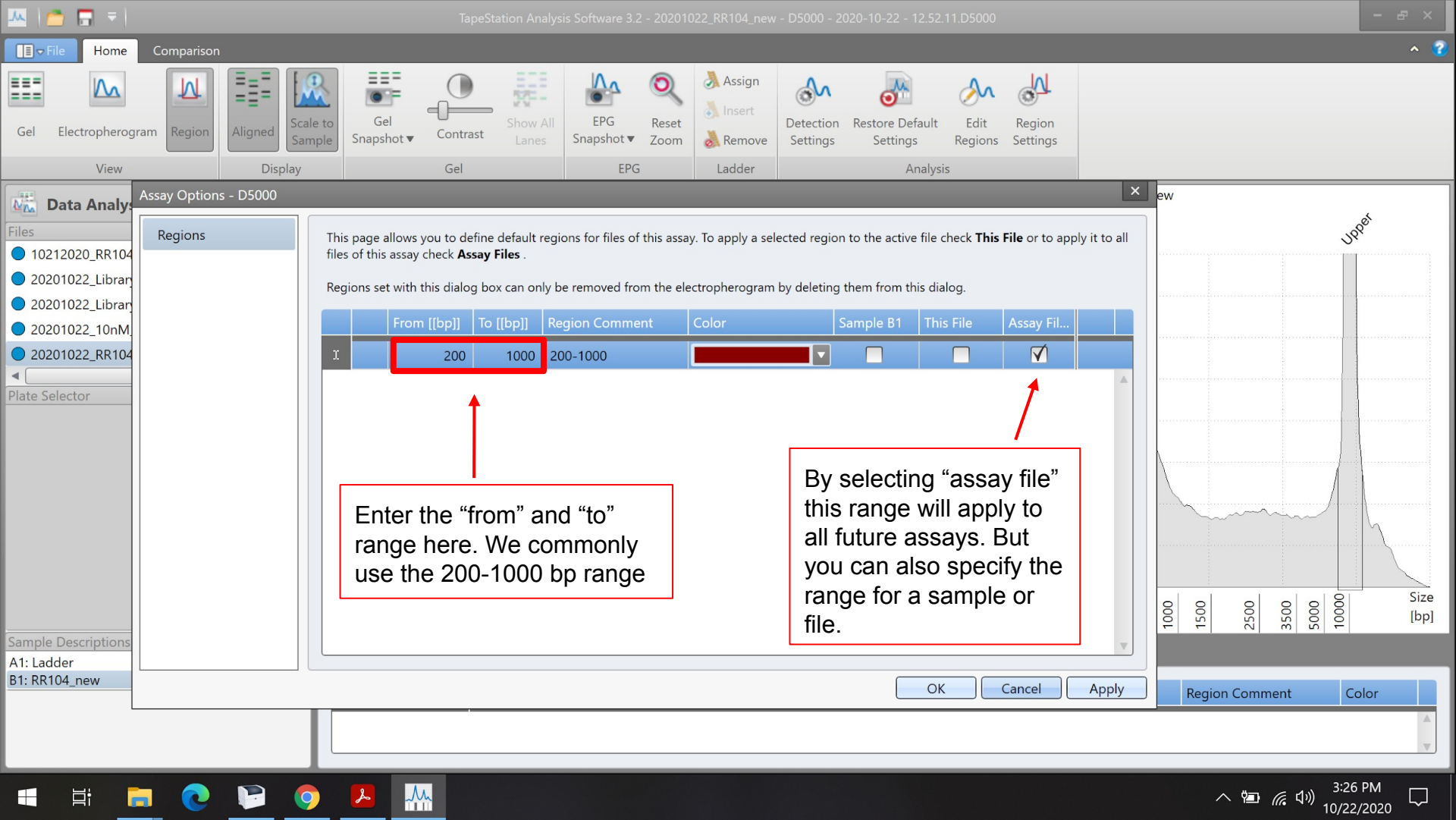


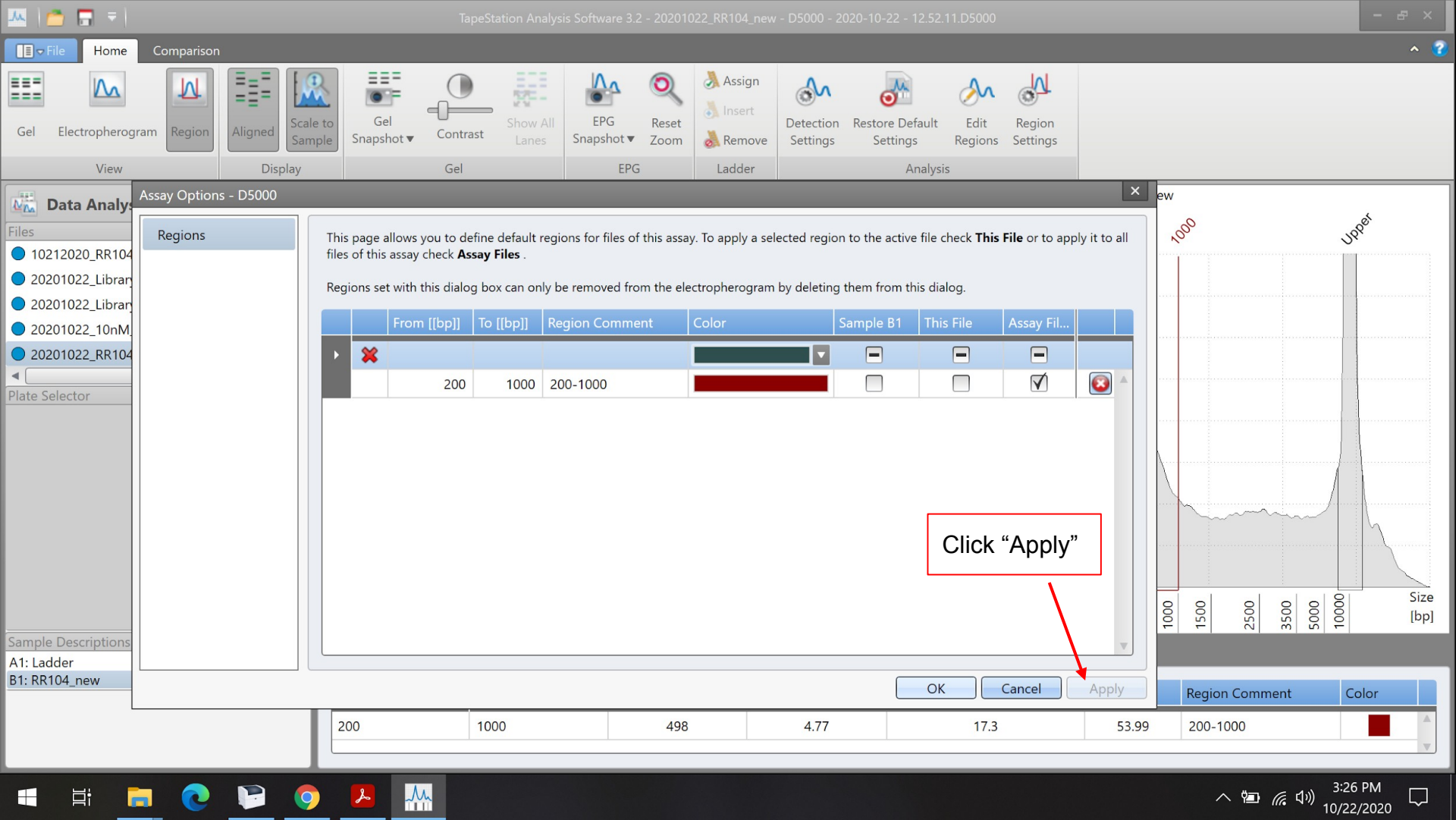
Region Table | Sample Table

From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color

3:22 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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10212020_RR104
20201022_Library
20201022_Library
20201022_10nM_
20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
Click here to add a new row
Click on this line to add the peak range of interest
OK
Cancel
Apply
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
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10/22/2020


TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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Electropherogram
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Aligned
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Detection Settings
Restore Default Settings
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Region Settings
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10212020_RR104
20201022_Library
20201022_Library
20201022_10nM
20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
200
1000
200-1000
Enter the “from” and “to” range here. We commonly use the 200-1000 bp range
By selecting “assay file” this range will apply to all future assays. But you can also specify the range for a sample or file.
OK
Cancel
Apply
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
3:26 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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20201022_RR104
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
Assay Options - D5000
Regions
This page allows you to define default regions for files of this assay. To apply a selected region to the active file check This File or to apply it to all files of this assay check Assay Files .
Regions set with this dialog box can only be removed from the electropherogram by deleting them from this dialog.
From [[bp]]
To [[bp]]
Region Comment
Color
Sample B1
This File
Assay Fil...
200
1000
200-1000
Click “Apply”
OK
Cancel
Apply
1000
Upper
1000
1500
2500
3500
5000
10000
Size [bp]
Region Comment
Color
200
1000
498
4.77
17.3
53.99
200-1000
3:26 PM
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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000

File | Home | Comparison

Gel | Electropherogram | Region | Aligned | Scale to Sample | Gel Snapshot | Contrast | Show All Lanes | EPG Snapshot | Reset Zoom | Assign | Insert | Remove | Detection Settings | Restore Default Settings | Edit Regions | Region Settings

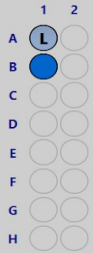
View | Display | Gel | EPG | Ladder | Analysis

Data Analysis

Files

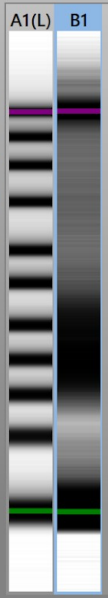
- 10212020_RR104_FS_DNAse_ARTIC - HSD5000
- 20201022_LibraryQC - D5000 - 2020-10-22 - 1
- 20201022_LibraryQC_HS - HSD5000 - 2020-10
- 20201022_10nM_PhiX_stocks - D5000 - 2020-1
- 20201022_RR104_new - D5000 - 2020-10-22 -

Plate Selector

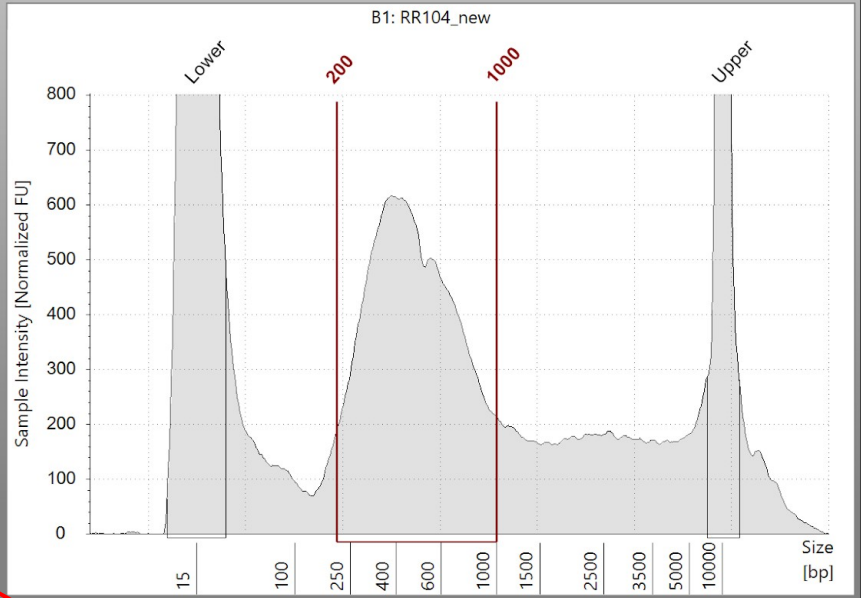


Sample Descriptions

A1: Ladder
B1: RR104_new



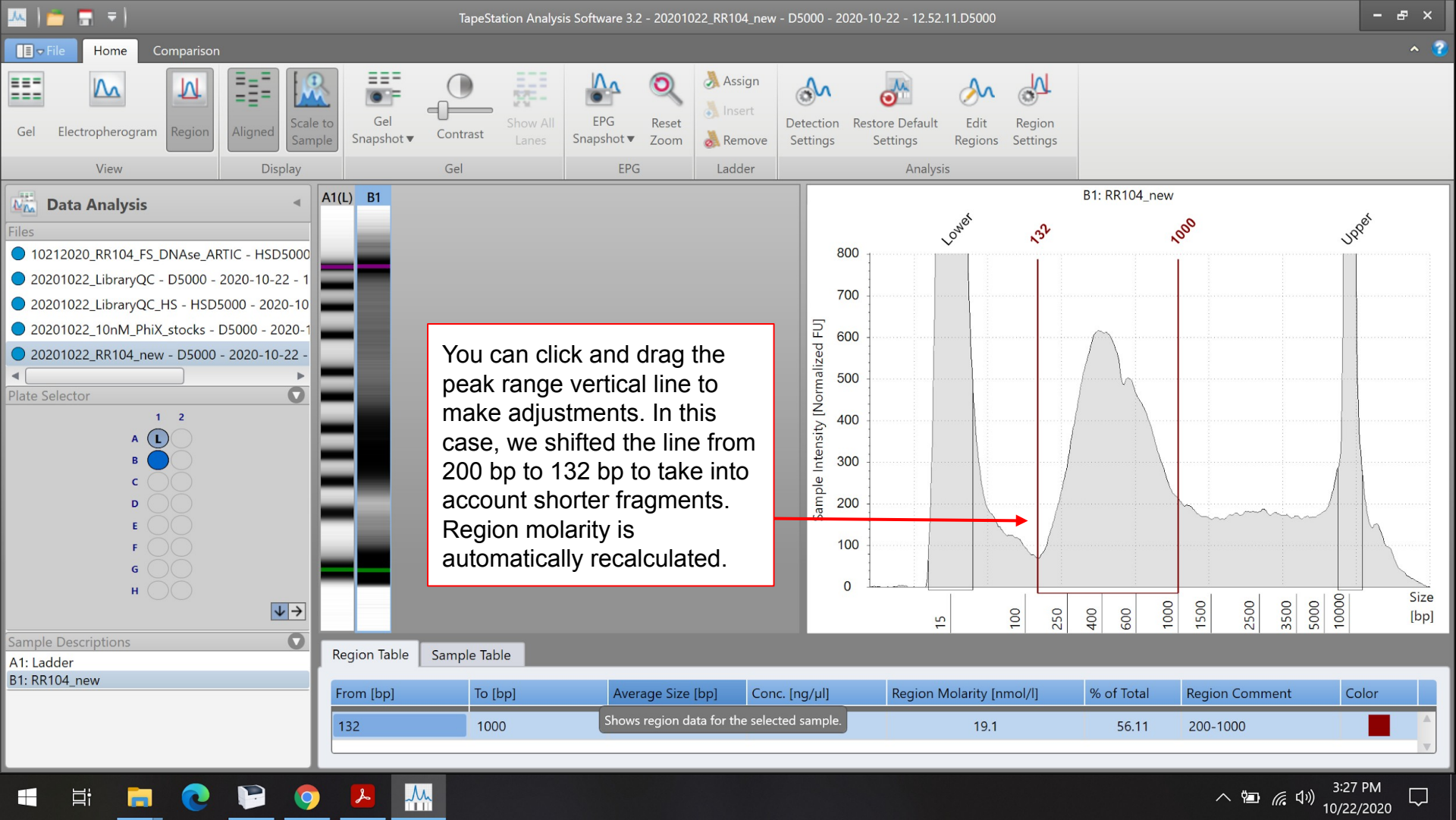
The “Region Molarity” concentration is now based on the 200-1000 bp range selected. This is the TS value that we use for our sample quantification

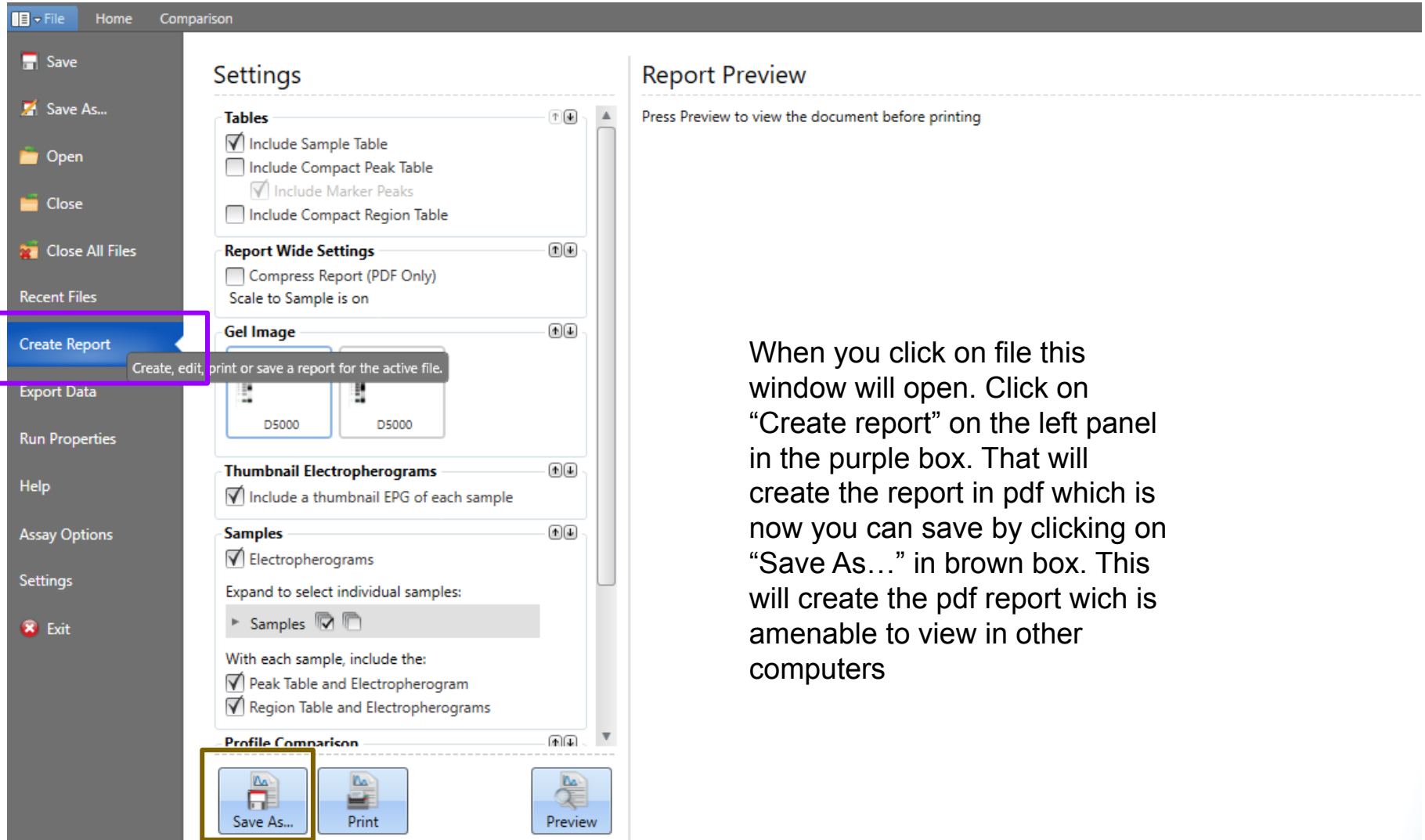


Region Table | Sample Table

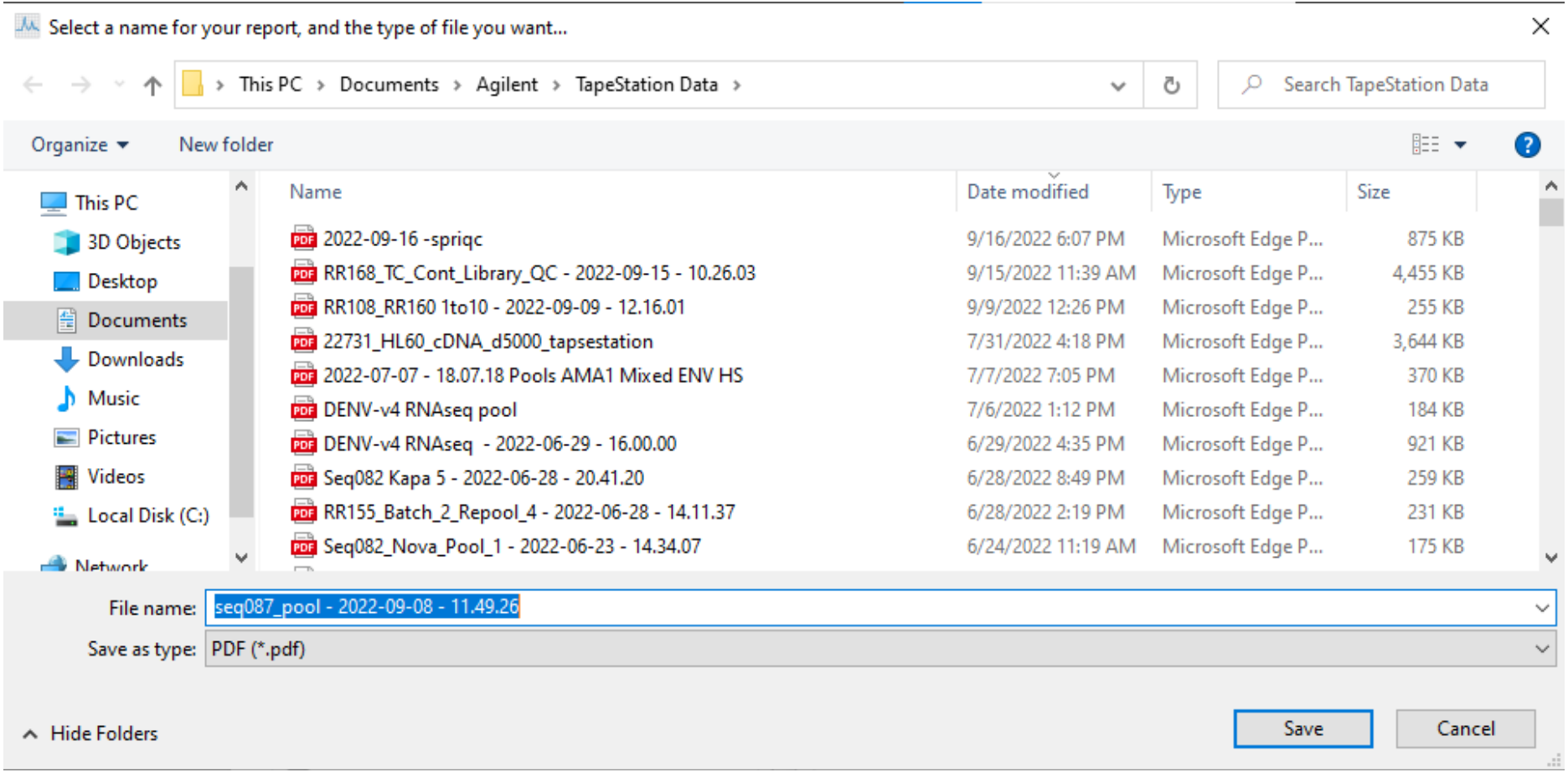
From [bp]	To [bp]	Average Size [bp]	Conc. [ng/µl]	Region Molarity [nmol/l]	% of Total	Region Comment	Color
200	1000	498	4.77	17.3	53.99	200-1000	

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TapeStation Analysis Software 3.2 - 20201022_RR104_new - D5000 - 2020-10-22 - 12.52.11.D5000
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20201022_LibraryQC_HS - HSD5000 - 2020-10
20201022_10nM_PhiX_stocks - D5000 - 2020-1
20201022_RR104_new - D5000 - 2020-10-22 -
Plate Selector
Sample Descriptions
A1: Ladder
B1: RR104_new
A1(L)
B1
You can click and drag the peak range vertical line to make adjustments. In this case, we shifted the line from 200 bp to 132 bp to take into account shorter fragments. Region molarity is automatically recalculated.
B1: RR104_new
Lower
132
1000
Upper
Sample Intensity [Normalized FU]
Size [bp]
Region Table
Sample Table
From [bp]
To [bp]
Average Size [bp]
Conc. [ng/µl]
Region Molarity [nmol/l]
% of Total
Region Comment
Color
132
1000
Shows region data for the selected sample.
19.1
56.11
200-1000
3:27 PM
10/22/2020



When you click on file this window will open. Click on “Create report” on the left panel in the purple box. That will create the report in pdf which is now you can save by clicking on “Save As…” in brown box. This will create the pdf report wich is amenable to view in other computers



Save and now your report is ready to share as a pdf!

Next: 06-O_Oxford Nanopore sequencing